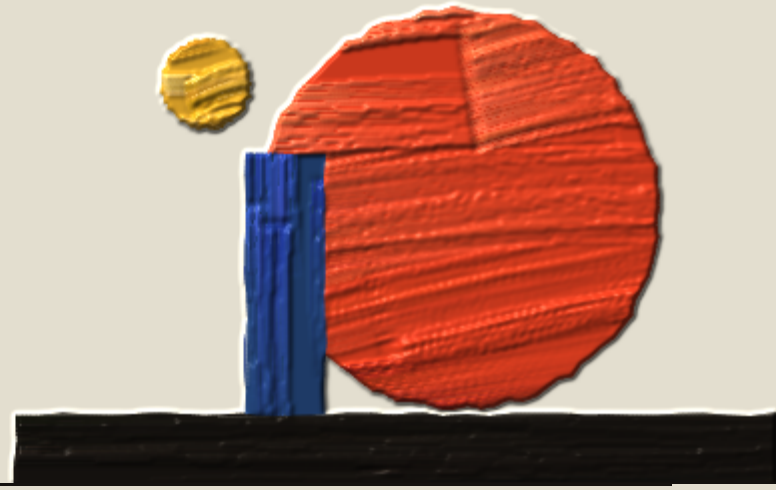
JOSE MANUEL ROJAS

JOSE MANUEL ROJAS



I don’t guess. I verify.

AI systems you can verify.

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LinkedIn GitHub

PROFILE

Seventh-semester Systems Engineering student at Universidad Icesi. I build agents, RAG and predictive models, and surround them with tests, guardrails and human review so they never guess. I use AI as a development team: I define the architecture, the contracts and the acceptance criteria, and the agents work inside their lane.

Looking for an internship and applied AI projects.

METHOD

HOW I WORK WITH AI

- **Contract first, code second.** Requirements and acceptance criteria are written before the first line.
- **Dangerous means blocked.** Push to main, force-push, mass deletes and reading .env are blocked, not politely requested.
- **Two failures and a person steps in.** After two failures the agent investigates; on the third it hands off to a human. Nothing ships without my OK.
- **Every agent in its lane.** A guard blocks edits outside the agent paths, including in parallel branches.
- **Tested or it is not done.** Tests, types, lint and secrets before anything counts as done. Then parallel review.
- **Memory in layers.** Short term: session events. Medium: status pinned in the issue. Long: ADRs and lessons.

PROJECTS

AgentSprint: a sprint board that gives coding agents memory

42

MCP TOOLS

PUBLISHED ON NPM, V0.2.1

Open source, MIT license

A sprint board that lives in Git and gives coding agents memory: every session starts with a summary of where the project stands and ends with a handoff for the next one.

- MCP server with 42 tools and 7 prompts: every Claude Code session receives the project brief (active sprint, what waits for review, what is next, blockers and lessons) and closes with a handoff in a journal.
- Review gate: read-only reviewers judge every task and send it back with findings when it is not right, so you only look at work that already passed. You can also approve from the chat.
- Monorepo of 5 TypeScript packages (core, CLI, MCP, server and a React web app) with 46 test files, continuous integration and published documentation.

TypeScript, Node.js, MCP, React, Fastify, Zod, Vitest, GitHub Actions

Multimodal RAG: retrieval and fusion across modalities

0.89

RECALL@1 TEXT TO IMAGE

RESEARCH

Artificial Intelligence course, 2026

An experimental study of how to retrieve and combine evidence from text, images, audio and tables in a RAG system, reporting results and limits without gloss.

- Text to image with CLIP ViT-B/32 on Flickr30k: Recall@1 of 0.89 and MRR of 0.936 over 100 queries.
- Text to audio with CLAP on AudioCaps: Recall@10 of 0.98.
- Tables: adding structure (company, region, frequency) raised Recall@1 from 0.67 to 1.00 compared with TF-IDF on a controlled corpus.

The benchmarks are small and partly synthetic, and the work says so.

Python, sentence-transformers, CLAP, scikit-learn, Colab

Three-level memory and MCP servers for agents

3

MEMORY LEVELS

COURSE CHALLENGE

Artificial Intelligence course, 2026

A manager that gives an assistant memory at three levels, a RAG over PDF evaluated with RAGAS and two MCP servers to connect tools to an agent.

- Three memory levels coordinated by a manager: short term (the last 10 messages), medium term (a session summary valid for 14 days) and long term (durable facts with importance, recency and a "right to be forgotten").
- Two MCP servers in Python (FastMCP): one for notes with 3 tools and another that searches and reads the official LangChain documentation from its llms.txt index.

The evaluation is small: 5 questions over a single document.

Python, Claude, ChromaDB, MCP (FastMCP), RAGAS, sentence-transformers, LangSmith



4 to 20x

CHEAPER THE SINGLE AGENT

RESEARCH

AI course experiment, 2026

Measured multi-agent architecture: is a team of agents worth it?

An agent system (orchestrator, parallel researchers, writer and critic) compared against a single agent with real models, measuring quality, cost, latency and coordination cost.

- The single agent turned out 4 to 20 times cheaper and 2.5 to 6 times faster, with equal or better quality. Coordination takes 21% to 40% of the cost.
- I measured the noise: the order between single agent and multi-agent flips between the two implementations, so I only call the differences in cost, latency and number of calls firm.

Starts from the course base notebooks, run with real models, fixed and documented.

Python, LangGraph, OpenRouter, DeepSeek, Claude (juez), Pydantic, Jupyter



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MODELS COMPARED

IN DEVELOPMENT

AI challenge with a healthcare company

Winback: reactivation prediction and WhatsApp chatbot

A model that estimates the probability that a former member returns within 12 months, to decide whom to contact first, plus a chatbot that talks and hands off to an advisor anyone who wants to come back.

- 12 models compared with bootstrap confidence intervals. I chose XGBoost for simplicity, because the ensembles tied within the error.
- Shuffled labels to rule out leakage, probability calibration and an internal audit with 10 faults fixed.
- Chatbot with rules for opt-out, complaint and advisor requests (no LLM calls); the model never sees the probability or health data. Sending is simulated, not tested with real WhatsApp.

Python, XGBoost, scikit-learn, FastAPI, Next.js, Supabase, pgvector



233,280

PROFILES VERIFIED

DEPLOYED PROTOTYPE

Mobile training app (PWA)

Rytm: hybrid lifting and running coach

An iPhone-installable PWA that guides the lifting session set by set, times cardio, builds a tailored plan and adds an AI coach and a social feed.

- Multi-provider LLM coach (Llama 3.3 on Workers AI by default; Gemini, Groq or Claude optional) that answers in JSON blocks validated by code.
- Deterministic plan engine, no AI: 39 lifting skeletons, 12 cardio templates and 700 tagged exercises, with invariants verified over 233,280 profile combinations.

No users yet.

TypeScript, React 19, Vite, Cloudflare Workers, D1, Durable Objects, Workers AI



15

AGENTS WITH THEIR OWN LANE

COMING SOON

With Daniel Escobar

Shift OS and Panel: AI agent teams with human approval

A system for organizing work teams that use AI: 15 agents with their own lane, versioned rules and a panel that shows in real time what each one is doing and lets you approve from the web or over WhatsApp.

- Versioned Claude Code plugin with 15 agents, 8 skills, hooks and a working constitution.
- My part: co-development, PR reviews and the idea of taking it to work teams as a product, which is the next step.

Claude Code, Python, Astro, React, Supabase, Cloudflare Workers, n8n

EXPERIENCE

Shift Digital 2026 - PRESENT

Co-founder. A launch-stage agency: websites and AI automation for small businesses in Colombia, the United States and Spain.

- Designed and built aishiftdigital.com, in production, with lead capture, a private inbox and an Instagram bot.
- Own products: WhatsApp CRM, customer-service agent with Claude, Mostrador and Shift OS.

EDUCATION

Universidad Icesi IN PROGRESS, SEVENTH SEMESTER

Systems Engineering. Artificial Intelligence (challenges with industry mentoring) and Internet Computing III.

STACK

AI AND DATA
Claude (Haiku, Sonnet, Opus), OpenAI-compatible APIs (Groq), Workers AI and Gemini, Hybrid RAG (full text and pgvector), ChromaDB y RAGAS, CLIP and CLAP, scikit-learn, XGBoost, LightGBM, Prompts with structured output

LANGUAGES
TypeScript, Python, SQL, JavaScript

DATA AND INFRASTRUCTURE
PostgreSQL and Supabase (RLS), Cloudflare (Workers, D1, R2), Vercel and Netlify, Docker

VIDEO AND AUDIO
GSAP, Remotion, ffmpeg, faster-whisper, Chatterbox

AGENTS
Claude Code (hooks, skills, subagents), LangGraph, MCP (servers in Python), n8n, OpenCode, AgentSprint (npm)

WEB AND BACKEND
React 19, Next.js, Astro, NestJS, FastAPI, Tailwind

QUALITY
Vitest, Playwright, pgTAP, GitHub Actions

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